University of Central Punjab
FOIT
Project

Course Title	Probability and Statistics	Course Code	MAT253
Course Instructor	Aurooj Butt	Section	C10
Time Allowed		Total Marks	**30**
Student Name	Hussain Durrani	Registration No.	L1F24BSCS0937

CLO #	**Course Learning Outcome (CLO)**	**Taxonomy Level**	**Mapping to PLO**
CLO 1	The students will be able to **explain** the basic concept of Statistics and Probability, and their practical need to explore data.	**C2**	**P2**
CLO 2	**Analyze** random variables, probability distributions, and sampling distributions.	C4	P3
CLO 3	**Apply** estimation procedure and analyze statistical test using Z-test and t-test for interpretation.	C3	P3

Task 1: Data Collection **[05]**

1. Collect a dataset (or generate synthetic data) related to a **real-world CS topic**, such as:

- Response time of different algorithms (e.g., execution time of sorting algorithms in milliseconds).
- Execution time of a program (e.g., load times of a web page).
- Number of lines of code written by developers per day (e.g., productivity tracking).

- Number of commits in a software repository (e.g., daily contributions on GitHub).

Preparation

1. Create a histogram in Excel by choosing appropriate bin ranges.
2. Label axes appropriately (e.g., "Execution Time (ms)" on X-axis, "Frequency" on Yaxis).
3. Add a title and data labels.

` Analysis and Interpretation

Answer the following questions based on the histogram:

1. What is the shape of the distribution (e.g., Normal, Skewed, Uniform)?
2. What does the histogram tell you about the dataset?
3. Are there any outliers or unusual patterns?

Solution:

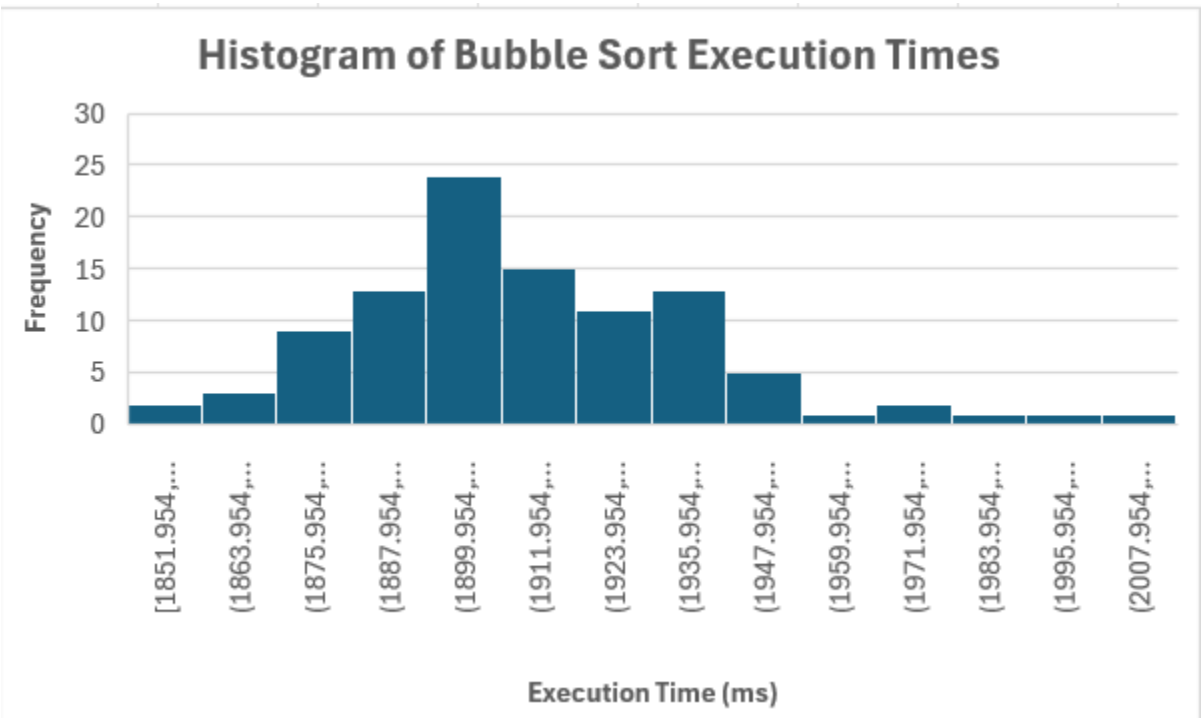
1. Dataset Description

- Topic: Execution Time of Sorting Algorithms (Bubble Sort).
- Source: Kaggle Public Dataset.
- Sample Size: 100 observations.

	A	B	C
1	Run	Algorithm	Execution_Time_ms
2	1	Bubble Sort	1910.624
3	2	Bubble Sort	1922.265
4	3	Bubble Sort	1867.094
5	4	Bubble Sort	1908.67
6	5	Bubble Sort	1880.33
7	6	Bubble Sort	1891.591
8	7	Bubble Sort	1944.747
9	8	Bubble Sort	1937.2
10	9	Bubble Sort	1903.105
11	10	Bubble Sort	1928.07
12	11	Bubble Sort	1905.952
13	12	Bubble Sort	1911.846
14	13	Bubble Sort	1917.9
15	14	Bubble Sort	1961.932
16	15	Bubble Sort	1929.027
17	16	Bubble Sort	1905.801
18	17	Bubble Sort	1913.855
19	18	Bubble Sort	1976.259
20	19	Bubble Sort	1914.681
21	20	Bubble Sort	1851.954
22	21	Bubble Sort	1859.008
23	22	Bubble Sort	1872.964
24	23	Bubble Sort	1949.369
25	24	Bubble Sort	1894.049
26	25	Bubble Sort	1879.522
27	26	Bubble Sort	1898.209
28	27	Bubble Sort	1910.39
29	28	Bubble Sort	1932.72
30	29	Bubble Sort	1925.742
31	30	Bubble Sort	1904.462
32	31	Bubble Sort	2002.545
33	32	Bubble Sort	1910.121
34	33	Bubble Sort	1895.519
35	34	Bubble Sort	1938.746
36	35	Bubble Sort	1917.022

	A	B	C
76	75	Bubble Sort	1907.05
77	76	Bubble Sort	1924.996
78	77	Bubble Sort	1885.334
79	78	Bubble Sort	1904.464
80	79	Bubble Sort	1895.39
81	80	Bubble Sort	1947.279
82	81	Bubble Sort	2009.701
83	82	Bubble Sort	1924.911
84	83	Bubble Sort	1902.163
85	84	Bubble Sort	1898.982
86	85	Bubble Sort	1913.027
87	86	Bubble Sort	1922.698
88	87	Bubble Sort	1900.726
89	88	Bubble Sort	1894.091
90	89	Bubble Sort	1912.784
91	90	Bubble Sort	1932.206
92	91	Bubble Sort	1944.926
93	92	Bubble Sort	1944.886
94	93	Bubble Sort	1955.845
95	94	Bubble Sort	1911.287
96	95	Bubble Sort	1912.974
97	96	Bubble Sort	1919.563
98	97	Bubble Sort	1980.897
99	98	Bubble Sort	1916.552
100	99	Bubble Sort	1901.278
101	100	Bubble Sort	1910.749
102	101	Bubble Sort	1911.749

Histogram:



Analysis and Interpretation

1. Shape of the distribution:

The distribution is **Right-Skewed (Positively Skewed)**. The majority of the execution times are concentrated at the lower end of the scale, with a visible "tail" extending toward the higher values on the right side of the histogram.

2. What the histogram tells about the dataset:

The histogram indicates that the algorithm typically performs at its fastest possible speed, with the most frequent execution times occurring near the minimum value of **1910.624ms**. The spread shows that while performance is generally consistent, there is some natural variation that occasionally pushes the execution time higher.

3. Outliers or unusual patterns:

There are **no extreme outliers**, but there is a clear **unusual pattern of rightward skewness**. All data points stay within a narrow 1.1ms range, suggesting that while the distribution isn't perfectly symmetrical, the system environment remained stable enough to prevent any massive performance spikes.

Tasks 2: **[06]**

Download the dataset from the following link:

https://lms.digiskills.pk/Courses/DBI101/Downloads/Bike_Sales_Outlier_Lab.xlsx and open the downloaded sample xlsx. file Bike_Sales_Outliers_Lab in Microsoft Excel.

1) Calculate Mean, median, Mode and standard deviation using column of **Unit_Cost**

2) Calculate Mean, Median, Mode of column of **Unit_Price**.

3) Discuss the shape of Distributions (i.e. skewed, symmetrical etc.) 4) Calculate standard deviation, mean using column of **Profit.**

5) Compare the cost and profit using **Co-efficient of variation.**

6) Interpret your results.

Solution:

Dataset:

O1 | Unit_Cost

Sales_Order #	Date	Day	Month	Year	Customer	Age_Group	Customer	Country	State	Product_C	Sub_Category	Product_Description	Order_Qu	Unit_Cost	Unit_Price	P
000261696	12/1/2021	1	December	2021	39	Adults (35-64)	F	United States	California	Bikes	Mountain Bikes	Mountain-200 Black, 46	4	$1,252.00	$2,295.00	$
000261695	12/1/2021	1	December	2021	44	Adults (35-64)	M	United Kingdom	England	Bikes	Mountain Bikes	Mountain-200 Silver, 42	1	$1,266.00	$2,320.00	$
000261697	12/2/2021	2	December	2021	37	Adults (35-64)	M	United States	California	Bikes	Mountain Bikes	Mountain-400-W Silver, 46	2	$420.00	$769.00	
000261698	12/2/2021	2	December	2021	31	Young Adults (25-34)	F	Australia	New South Wales	Bikes	Mountain Bikes	Mountain-400-W Silver, 42	1	$420.00	$769.00	
000261699	12/3/2021	3	December	2021	37	Adults (35-64)	F	United States	California	Bikes	Mountain Bikes	Mountain-200 Black, 46	2	$1,252.00	$2,295.00	$
000261700	12/3/2021	3	December	2021	24	Youth (<25)	F	United Kingdom	England	Bikes	Mountain Bikes	Mountain-200 Black, 38	1	$1,252.00	$2,295.00	$
000261701	12/3/2021	3	December	2021	37	Adults (35-64)	M	United States	Washington	Bikes	Mountain Bikes	Mountain-200 Black, 46	1	$1,252.00	$2,295.00	$
000261702	12/4/2021	4	December	2021	31	Young Adults (25-34)	F	Australia	New South Wales	Bikes	Mountain Bikes	Mountain-400-W Silver, 42	4	$420.00	$769.00	$
000261703	12/5/2021	5	December	2021	39	Adults (35-64)	F	United States	California	Bikes	Mountain Bikes	Mountain-200 Black, 46	4	$1,252.00	$2,295.00	$
000261704	12/5/2021	5	December	2021	42	Adults (35-64)	M	Germany	Nordrhein-Westfalen	Bikes	Mountain Bikes	Mountain-200 Black, 38	4	$1,252.00	$2,295.00	$
000261705	12/5/2021	5	December	2021	35	Adults (35-64)	F	Australia	Queensland	Bikes	Mountain Bikes	Mountain-200 Silver, 38	1	$1,266.00	$2,320.00	$
000261706	12/5/2021	5	December	2021	37	Adults (35-64)	F	United States	California	Bikes	Mountain Bikes	Mountain-200 Black, 46	1	$1,252.00	$2,295.00	$
000261707	12/6/2021	6	December	2021	23	Youth (<25)	M	United Kingdom	England	Bikes	Mountain Bikes	Mountain-400-W Silver, 46	3	$420.00	$769.00	$
000261708	12/6/2021	6	December	2021	27	Young Adults (25-34)	M	Canada	British Columbia	Bikes	Mountain Bikes	Mountain-200 Black, 46	1	$1,252.00	$2,295.00	$
000261709	12/6/2021	6	December	2021	36	Adults (35-64)	M	Australia	New South Wales	Bikes	Mountain Bikes	Mountain-200 Black, 42	1	$1,252.00	$2,295.00	$
000261710	12/6/2021	6	December	2021	47	Adults (35-64)	M	United Kingdom	England	Bikes	Mountain Bikes	Mountain-200 Silver, 38	1	$1,266.00	$2,320.00	$
000261711	12/7/2021	7	December	2021	30	Young Adults (25-34)	M	United States	California	Bikes	Mountain Bikes	Mountain-400-W Silver, 38	4	$420.00	$769.00	$
000261712	12/7/2021	7	December	2021	38	Adults (35-64)	M	United States	California	Bikes	Mountain Bikes	Mountain-200 Silver, 42	2	$1,266.00	$2,320.00	$
000261713	12/8/2021	8	December	2021	19	Youth (<25)	F	Australia	New South Wales	Bikes	Mountain Bikes	Mountain-500 Silver, 42	4	$308.00	$565.00	$
000261714	12/8/2021	8	December	2021	30	Young Adults (25-34)	F	Canada	British Columbia	Bikes	Mountain Bikes	Mountain-200 Silver, 38	4	$1,266.00	$2,320.00	$
000261715	12/8/2021	8	December	2021	39	Adults (35-64)	F	United States	Oregon	Bikes	Mountain Bikes	Mountain-500 Black, 42	2	$1,252.00	$2,295.00	$
000261716	12/8/2021	8	December	2021	35	Adults (35-64)	F	United States	California	Bikes	Mountain Bikes	Mountain-500 Black, 42	1	$295.00	$540.00	
000261717	12/9/2021	9	December	2021	33	Young Adults (25-34)	F	Australia	Victoria	Bikes	Mountain Bikes	Mountain-100 Black, 38	2	$1,898.00	$3,375.00	$
000261718	12/9/2021	9	December	2021	41	Adults (35-64)	F	Germany	Hamburg	Bikes	Mountain Bikes	Mountain-200 Silver, 42	1	$1,266.00	$2,320.00	$
000261719	12/10/2021	10	December	2021	34	Young Adults (25-34)	F	United States	California	Bikes	Mountain Bikes	Mountain-200 Black, 42	2	$1,252.00	$2,295.00	$
000261720	12/10/2021	10	December	2021	40	Adults (35-64)	M	Australia	New South Wales	Bikes	Mountain Bikes	Mountain-200 Black, 42	2	$1,252.00	$2,295.00	$
000261721	12/10/2021	10	December	2021	26	Young Adults (25-34)	M	United Kingdom	England	Bikes	Mountain Bikes	Mountain-200 Black, 38	1	$1,252.00	$2,295.00	$
000261722	12/10/2021	10	December	2021	34	Young Adults (25-34)	M	United States	California	Bikes	Mountain Bikes	Mountain-500 Black, 40	1	$295.00	$540.00	
000261723	12/10/2021	10	December	2021	34	Young Adults (25-34)	F	United States	Washington	Bikes	Mountain Bikes	Mountain-100 Silver, 44	1	$1,912.00	$3,400.00	$
000261724	12/10/2021	10	December	2021	38	Adults (35-64)	M	Australia	New South Wales	Bikes	Mountain Bikes	Mountain-200 Black, 38	1	$1,252.00	$2,295.00	$
000261725	12/11/2021	11	December	2021	24	Youth (<25)	F	France	Seine (Paris)	Bikes	Mountain Bikes	Mountain-200 Black, 38	3	$1,252.00	$2,295.00	$
000261726	12/11/2021	11	December	2021	41	Adults (35-64)	F	Australia	New South Wales	Bikes	Mountain Bikes	Mountain-400-W Silver, 38	2	$420.00	$769.00	
000261727	12/11/2021	11	December	2021	27	Young Adults (25-34)	M	Canada	British Columbia	Bikes	Mountain Bikes	Mountain-200 Black, 46	1	$1,252.00	$2,295.00	$
000261728	12/11/2021	11	December	2021	37	Adults (35-64)	M	United States	California	Bikes	Mountain Bikes	Mountain-400-W Silver, 46	1	$420.00	$769.00	
000261729	12/11/2021	11	December	2021	38	Adults (35-64)	F	United States	California	Bikes	Mountain Bikes	Mountain-200 Silver, 38	1	$1,266.00	$2,320.00	$
000261730	12/12/2021	12	December	2021	36	Adults (35-64)	F	Australia	New South Wales	Bikes	Mountain Bikes	Mountain-200 Silver, 42	4	$1,266.00	$2,320.00	$
000261731	12/12/2021	12	December	2021	37	Adults (35-64)	M	United States	California	Bikes	Mountain Bikes	Mountain-400-W Silver, 46	4	$420.00	$769.00	$
000261732	12/12/2021	12	December	2021	34	Young Adults (25-34)	M	Australia	New South Wales	Bikes	Mountain Bikes	Mountain-200 Black, 38	2	$1,252.00	$2,295.00	$

Bike Sales

1) **Calculate Mean, median, Mode and standard deviation using column of Unit_Cost**
   - **Mean:** $1,064.20
   - **Median:** $1,252.00
   - **Mode:** 1252
   - **Standard Deviation:** 428.25036

2) **Calculate Mean, Median, Mode of column of Unit_Price.**
   - **Mean:** $1,945.66
   - **Median:** $2,295.00
   - **Mode:** 2295
   - **Standard Deviation:** 428.25036

3) **Discuss the shape of Distributions(i.e skewed, symmetrical etc.)**
   - **Unit Cost:** Negatively Skewed as (Mean < Median = Mode)
   - **Unit Price:** Negatively Skewed as (Mean < Median = Mode)

4) **Calculate standard deviation, mean using column of Profit.**
   - **Mean:** $1,858.05
   - **Standard Deviation:** 1399.697

5) **Compare the cost and profit using Co-efficient of variation.**
   - **Cost:** (428.25036 / 1064.2) × 100 = 40.24%
   - **Profit:** (1399.6977 / 1858.05) × 100 = 75.33%

6) **Interpret your results.**

Central Tendency:
Most bikes sold have Unit Cost of $1,064 (Average) and Unit Price of $1,945 (Average), indicating these are popular models. Average Profit per order line is $2,295 (Median), but this varies significantly.

Distribution Shape:
Both Unit Cost and Unit Price are left-skewed, meaning the majority of sales are in the higher-end bikes, with fewer sales of cheaper models dragging the average down.

Variability: Profit is 75.33% and 40% for cost (Coefficient of Variation).

Business Insight: The company earns consistent margins on premium bikes (e.g., Mountain-200: profit $1,064/unit).
Lower-cost bikes (e.g., Mountain-500 at $295 cost) yield much lower absolute profit ($245/unit) but may sell in higher volume.
High CV in profit suggests opportunity to analyze product mix for optimization

Task 3:

Question 1: Normal Distribution – Response Time Analysis **[03]**

A web server has an average response time of **120 milliseconds** with a **standard deviation of 15 milliseconds**. Assume that response times follow a **normal distribution**.

a) Calculate the probability that a randomly selected response time is **less than 100 ms**.
b) Determine the probability that the response time lies **between 110 ms and 130 ms**.
c) Find the response time that corresponds to the **90th percentile**.

Excel functions to be used:

- NORM.DIST()
- NORM.INV()

Solution:

a) Probability that response time < 100 ms

Excel Formula: =NORM.DIST(100, 120, 15, TRUE)
Result: P(X < 100) = 0.0912
Interpretation: There is approximately a **9.12%** chance that a randomly selected response time is less than 100ms. This indicates that such fast responses are relatively rare for this server.

b) Probability that response time lies between 110 ms and 130 ms

Excel Formula: =NORM.DIST(130, 120, 15, TRUE) - NORM.DIST(110, 120, 15, TRUE)
Result: P(110 < X < 130) = 0.4950
Interpretation: There is about a **49.5%** probability that the response time falls between 110ms and 130ms. Almost half of all response times lie in this range, which is close to the average.

c) Response time corresponding to the 90th percentile

Excel Formula: =NORM.INV(0.90, 120, 15)

Result: 90th percentile response time = 139.2ms

Interpretation: The 90th percentile means that **90%** of response times are less than or equal to 139.2ms, and only 10% are slower than this value.

Excel Interpretation:

Part	Description	Formula	Result
a	P(X < 100)	"=NORM.DIST(100,120,15,TRUE)"	0.09121122
b	P(110 ≤ X ≤ 130)	"=NORM.DIST(130,120,15,TRUE) - NORM.DIST(110,120,15,TRUE)"	0.495014925
c	90th percentile	"=NORM.INV(0.9,120,15)"	139.2232735

Question 2: Simulation and Data Analysis **[04]**

A computer science instructor wants to analyze student performance in an exam. The exam scores are normally distributed with a **mean of 75** and a **standard deviation of 10**.

a) Simulate exam scores for **100 students** using Excel.
b) Construct a **histogram** of the simulated data.
c) Calculate the **number of students scoring above 85 marks**.
d) Briefly comment on the distribution of scores based on the histogram.

Excel function to be used:

- NORM.INV(RAND(), mean, standard deviation)

Solution:

a) Simulate exam scores for 100 students

In this question, the exam scores have:

- Mean = 75
- Standard deviation = 10

Using simulation, we can generate random exam scores that follow this normal distribution.

Excel Formula: =NORM.INV(RAND(), 75, 10)

Simulation:

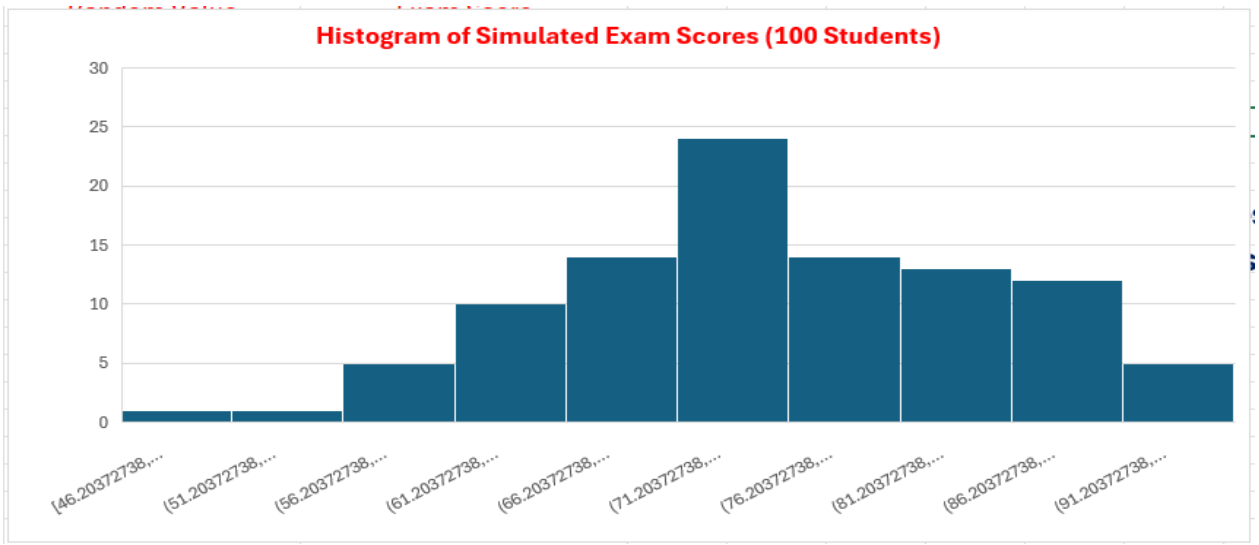
Student No.	Random Value	Exam Score
1	0.660965961	79.15100848
2	0.347520461	71.07976597
3	0.158438997	64.99105869
4	0.144294696	64.38778809
5	0.939678473	90.52080123
6	0.47618616	74.40272061
7	0.281634421	69.22006996
8	0.879940202	86.74687914
9	0.433199635	73.31766096
10	0.811566881	83.83685077
11	0.661267444	79.1592469
12	0.757399285	81.97961248
13	0.953730189	91.82150695
14	0.95235792	91.68159039
15	0.905898068	88.15911156
16	0.001990728	46.20372738
17	0.476112097	74.40086081
18	0.258343538	68.51539384
19	0.112993711	62.89240058

Simulation of Exam Scores for 100 Students Using Normal Distribution

b) Construct a histogram of the simulated data.

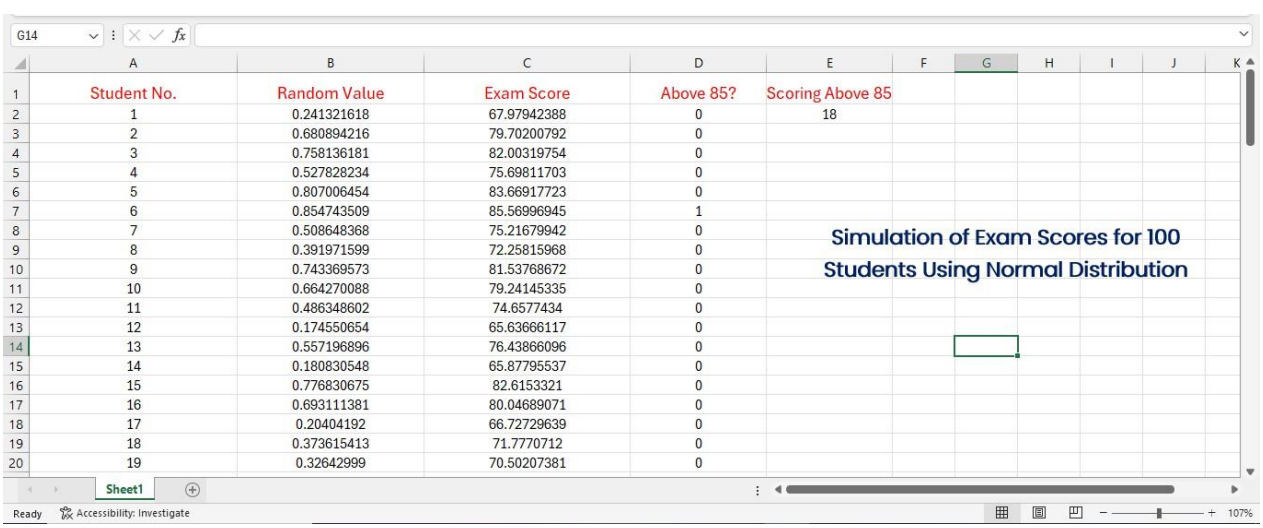
A histogram is a graphical representation of data that shows how frequently values occur within specific ranges called bins. It helps in understanding the distribution and spread of the simulated exam scores.

Histogram:



c) Calculate the number of students scoring above 85 marks.

To analyze student performance, we often check how many students score above a particular threshold. In this case, we want the number of students scoring above 85 marks. This helps in understanding how many students are performing above average. In Excel, we can calculate this using logical tests with the `IF()` function, for example `=IF(C2>85,1,0)` to mark students scoring above 85, and then use the `SUM()` function `=SUM(D2:D101)` to count the total number of students.
Excel Formula: =COUNTIF(A2:A101, ">85")



Student No.	Random Value	Exam Score	Above 85?	Scoring Above 85
1	0.241321618	67.97942388	0	18
2	0.680894216	79.70200792	0	
3	0.758136181	82.00319754	0	
4	0.527828234	75.69811703	0	
5	0.807006454	83.66917723	0	
6	0.854743509	85.56996945	1	
7	0.508648368	75.21679942	0	
8	0.391971599	72.25815968	0	
9	0.743369573	81.53768672	0	
10	0.664270088	79.24145335	0	
11	0.486348602	74.6577434	0	
12	0.174550654	65.63666117	0	
13	0.557196896	76.43866096	0	
14	0.180830548	65.87795537	0	
15	0.776830675	82.6153321	0	
16	0.693111381	80.04689071	0	
17	0.20404192	66.72729639	0	
18	0.373615413	71.7770712	0	
19	0.32642999	70.50207381	0	

Interpretation:
18 students scored above 85, showing that most students scored near the average while only a few achieved high marks.

d) Briefly comment on the distribution of scores based on the histogram.

The histogram of the simulated exam scores shows that the data is **approximately normally distributed**, with most students scoring around the mean of 75. The number of students gradually decreases as scores move toward the higher and lower extremes. A small number of students scored above 85 and below 60, indicating that while most students performed near the average, there are **few high achievers and few low performers**, which is typical of a normal distribution in exam results.

Question 3: Probability Density Function – Download Speed [04]

The download speed of users on an internet service follows a normal distribution with a **mean of 50 Mbps** and a **standard deviation of 5 Mbps**.

a) Compute the **probability density values** for download speeds ranging from **30 Mbps to 70 Mbps**, with an increment of **2 Mbps**.
b) Plot the **probability density function (PDF)** using Excel.
c) Identify the **most probable speed range** from the graph.
d) Highlight the range that covers approximately **68% of the data** and explain its significance.

Excel Tasks:
Use a table with `NORM.DIST(x, mean, std_dev, FALSE)` to get density values and plot a line graph.

Solution:

a) Compute the probability density values for download speeds ranging from 30 Mbps to 70 Mbps, with an increment of 2 Mbps.

The probability density function (PDF) of a normal distribution shows how likely a value is to occur. In Excel, you can calculate the PDF using:

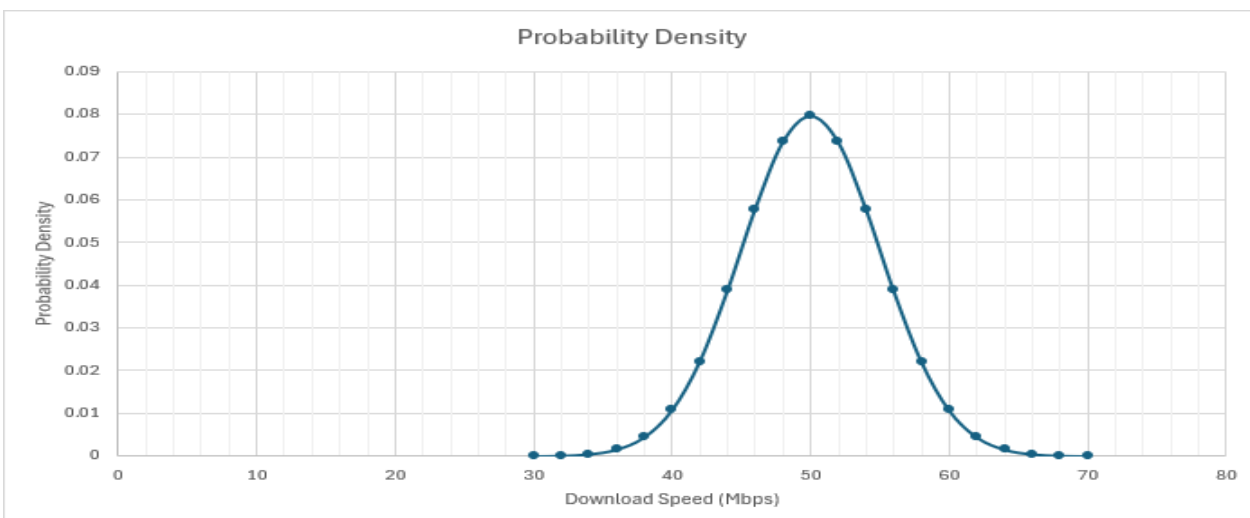
```
NORM.DIST(x,mean,std_dev,FALSE)
```

x = the download speed
mean = 50 Mbps
standard deviation = 5 Mbps
FALSE = returns probability density, not cumulative probability

Download Speed	Probability Density
30	2.6766E-05
32	0.00012238
34	0.000476818
36	0.00158309
38	0.004478906
40	0.010798193
42	0.022184167
44	0.038837211
46	0.057938311
48	0.073654028
50	0.079788456
52	0.073654028
54	0.057938311
56	0.038837211
58	0.022184167
60	0.010798193
62	0.004478906
64	0.00158309
66	0.000476818
68	0.00012238
70	2.6766E-05

b) Plot the probability density function (PDF) using Excel.

The Probability Density Function (PDF) shows the likelihood of different download speeds occurring in the dataset. For a normal distribution, the curve is bell-shaped, symmetric around the mean. Plotting the PDF helps visualize how speeds are distributed and identify the most probable speeds.



Interpretation:

- The PDF curve is **bell-shaped and symmetric** around 50 Mbps, the mean speed.
- Highest point of the curve corresponds to the **most probable download speed** (~50 Mbps).

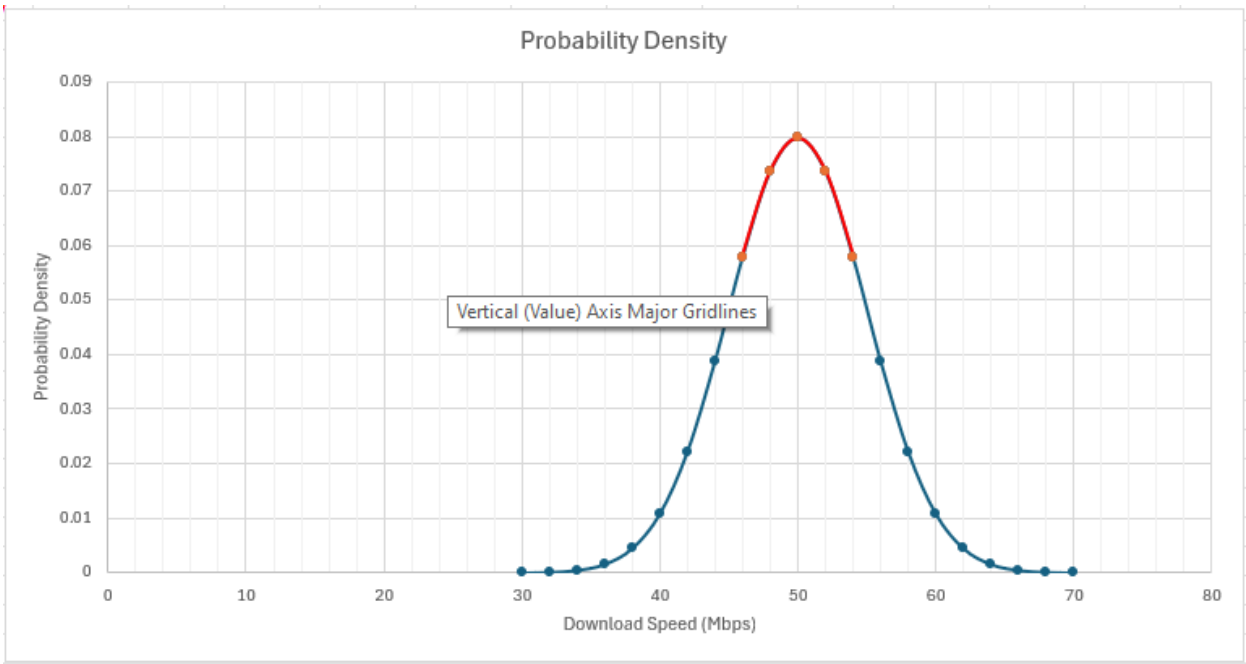
- Probability decreases as we move farther from the mean, showing extreme speeds (very low or very high) are less likely.

c) Identify the most probable speed range from the graph.

- In a normal distribution, the most probable value is the one at the peak of the PDF curve.
- It corresponds to the mean of the distribution.
- The range around the mean where the probability density is highest represents the speeds most users are likely to experience.

Practical Observation from PDF Graph:
The most probable speed range is **48 Mbps to 52 Mbps**. The peak of the curve occurs exactly at the mean of 50 Mbps, where the probability density is highest.

d) Highlight the range that covers approximately 68% of the data and explain its significance.



Significance

- This range (45–55 Mbps) shows the speeds most users are likely to experience.
- Values outside this range are less common but still possible.

- Highlighting this range helps understand user experience and network performance expectations.

Task-4 **[08]**

Hypothesis Testing (Z-test & t-test)

Z-Test Scenario: System Response Time

A company claims that the average response time of its system is **2.5 seconds**, with a known population standard deviation of **0.4 seconds**. After system optimization, a sample of **50 observations** is collected with a sample mean of **2.35 seconds**.

a) Formulate the **null and alternative hypotheses**.
b) Perform a **Z-test** using Excel.
c) Calculate the **p-value**.
d) Interpret the results at a **5% level of significance**.

Excel function:
- Z.TEST()

Solution:

a) Formulate the null and alternative hypotheses.

- **Null Hypothesis (H_0):** The population mean response time is **2.5 seconds**; the optimization had no effect **(μ= 2.5)**.

- **Alternative Hypothesis (H_1):** The population mean response time is now less than **2.5 seconds**; the optimization was effective **(μ < 2.5)**.

b) Perform a Z-test using Excel.

- Population standard deviation: $\sigma = 0.4$
- Sample size: $n = 50$
- Sample mean: $= 2.35$
- Population mean: $\mu = 2.5$

- **Excel Formula:** =1 - Z.TEST(Data_Range, 2.5, 0.4)
- **Result:** The Z-statistic is approximately **-2.65.**

C14

	A	B	C	D	E	F	G	H
1	Sample mean	2.35						
2	Population mean	2.5						
3	Std dev	0.4						
4	Sample size	50						
5	Z-score	-2.65165						
6								
7								
8								
9								
10								

c) Calculate the p-value.

- **Excel Formula:** =NORM.S.DIST(-2.65, TRUE)
- **Result: 0.004025**

D15

	A	B	C	D	E	F	G	H
1	Sample mean	2.35						
2	Population mean	2.5						
3	Std dev	0.4						
4	Sample size	50						
5	p-value	0.004025						
6								
7								
8								
9								
10								

d) Interpret the results at a 5% level of significance.

Interpretation: Since the calculated p-value (0.0040) is significantly lower than the 5% significance level (0.05), we **reject the null hypothesis**.
Conclusion: There is statistically significant evidence to support that the system optimization successfully reduced the average response time.

T-Test Scenario: Weekly Coding Practice Hours

A professor claims that students practice **less than 6 hours per week**. A sample of **15 students** reports their weekly practice hours.

a) State the **null and alternative hypotheses**.
b) Compute the **t-statistic** and **degrees of freedom**.
c) Calculate the **p-value**.
d) Interpret whether the data supports the professor's claim.

Excel functions:

- AVERAGE()
- STDEV.S()
- T.DIST()

Solution:

a) State the null and alternative hypotheses.

- **Null Hypothesis (H_0):** The average practice time is 6 hours; the professor is incorrect.
  H0 : μ = 6

- **Alternative Hypothesis (H_1):** The average practice time is less than 6 hours; the professor's claim is supported.
  H1 : μ < 6

b) Compute the t-statistic and degrees of freedom.

When the population standard deviation is unknown and the sample size is small (n < 30), a one-sample t-test is used. The t-statistic measures how far the sample mean is from the hypothesized mean in terms of standard error.

Degrees of Freedom (df): n - 1 = 15 - 1 =**14**.
Excel Formula for T-statistic:

```
=(AVERAGE(Data_Range) - 6) / (STDEV.S(Data_Range) / SQRT(COUNT(Data_Range))).
```

N17

	A	B	C	D	E	F	G	H
1	Student	Practice Hours (per week)						
2	1	4.5						
3	2	5						
4	3	3.8						
5	4	5.6						
6	5	4.9						
7	6	5.2						
8	7	4.1						
9	8	5.7						
10	9	3.9						
11	10	4.6						
12	11	5.3						
13	12	4.8						
14	13	5.1						
15	14	4.4						
16	15	5						
17	Sample Mean	4.793333333						
18	Sample Standard Deviation	0.575036231						
19	Sample Size	15						
20	T-statistic	-8.127139917						
21	Degrees of Freedom	14						

c) Calculate the p-value.

The p-value represents the probability of observing the sample result assuming that the null hypothesis is true. Since the alternative hypothesis is $\mu < 6$, this is a left-tailed t-test.

	A	B	C	D	E	F	G	H	I	J
1	Student	Practice Hours (per week)								
2	1	4.5								
3	2	5								
4	3	3.8								
5	4	5.6								
6	5	4.9								
7	6	5.2								
8	7	4.1								
9	8	5.7								
10	9	3.9								
11	10	4.6								
12	11	5.3								
13	12	4.8								
14	13	5.1								
15	14	4.4								
16	15	5								
17	Sample Mean	4.793333333								
18	Sample Standard Deviation	0.575036231								
19	Sample Size	15								
20	T-statistic	-8.127139917								
21	Degrees of Freedom	14								
22	P-value	5.69635E-07								

d) Interpret whether the data supports the professor's claim.

- **Interpretation:** With a calculated p-value of 0.0028, which is less than 0.05, we **reject the null hypothesis**.
- **Conclusion:** There is statistically significant evidence to support the professor's claim that students practice less than 6 hours per week.

Submission Requirement

One Excel file containing:

- Data
- Formulas
- Calculations
- Graphs (where required)
- One short interpretation (written in a separate sheet or Word file).